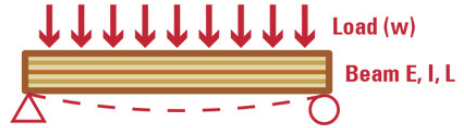
MPP

MASS PLY BY FRERES

ROOF TABLE NOTES:

- Single Span, uniform load.
- Total loads include live load, Mass Ply weight, and 15 PSF dead load.
- Deflection limits are for live loads, Total load deflection limits for live load deflection of L/360, L/240, and L/180 are respectively L/240, L/180, and L/120
- Tables do not substitute for the work of a design professional.



F10 MASS PLY MAXIMUM SPAN ROOF TABLE - MAJOR FORCE DIRECTION

		SNOW LOAD, UNFACTORED															125% NON-SNOW LOAD		
LAY-UP	THICKNESS	20 PSF LIVE LOAD			30 PSF LIVE LOAD			40 PSF LIVE LOAD			55 PSF LIVE LOAD			100 PSF LIVE LOAD			20 PSF LIVE LOAD		
ID	(IN)	L/360	L/240	L/180	360	240	180	360	240	180	360	240	180	360	240	180	360	240	180
F10-2	2-1/16	7.28	8.02	9.19	6.76	7.45	8.54	6.36	7.01	8.04	5.74	6.51	7.27	4.68	5.38	5.93	7.28	8.02	9.19
F10-3	3-1/16	10.70	11.79	13.52	9.98	11.01	12.62	9.43	10.39	11.92	8.66	9.68	10.96	7.05	8.11	8.94	10.70	11.79	13.52
F10-4	4-1/16	13.84	15.25	17.49	12.97	14.30	16.40	12.28	13.54	15.53	11.46	12.64	14.50	9.33	10.73	11.84	13.84	15.25	17.49
F10-5	5-1/8	16.96	18.70	21.44	15.95	17.59	20.18	15.14	16.70	19.16	14.18	15.64	17.95	11.69	13.44	14.83	16.96	18.70	21.44
F10-6	6-1/8	19.95	21.99	25.22	18.82	20.75	23.80	17.90	19.75	22.66	16.80	18.54	21.28	14.02	16.11	17.79	19.95	21.99	25.22
F10-7	7-1/8	22.81	25.15	28.85	21.58	23.80	27.31	20.58	22.70	26.05	19.36	21.36	24.52	16.35	18.64	20.74	22.81	25.15	28.85
F10-8	8-3/16	25.61	28.23	32.38	24.29	26.79	30.74	23.21	25.60	29.38	21.88	24.14	27.71	18.68	21.14	23.71	25.61	28.23	32.38
F10-9	9-3/16	28.32	31.23	35.83	26.93	29.71	34.08	25.78	28.44	32.63	24.35	26.87	30.85	21.03	23.61	26.68	28.32	31.23	35.83
F10-10	10-3/16	30.95	34.13	39.15	29.50	32.53	37.33	28.28	31.90	35.80	26.76	29.53	33.90	23.36	26.03	29.64	30.95	34.13	39.15
F10-11	11-1/4	33.52	36.97	42.41	32.01	35.31	40.52	30.74	33.91	38.92	29.14	32.16	36.92	25.70	28.43	32.60	33.52	36.97	42.41
F10-12	12-1/4	36.02	39.73	45.58	34.46	38.02	43.62	33.14	36.56	41.96	31.47	34.73	39.87	27.89	30.80	35.38	36.02	39.73	45.58

SPAN & LOAD TABLE…from Freres Engineered Wood (use this Table for Wood Building)

GLULAM Purlins

(use this Table for Wood Building)

3½" and 5½" Roof Load Tables – 115%

Span [ft]	Span Type	24F-V4 Grade – 115% Load Duration In pounds per lineal foot (PLF)									
		3½"					5½"				
		7½"	9½"	11⅞"	14"	16"	7½"	9½"	11⅞"	14"	16"
6	Simple	1671	2683	4190	5418	6813	2625	4216	6584	8514	10706
	Multiple	1286	2066	3049	3842	4694	2021	3247	4791	6037	7377
	Min. Bearing	2.2 / 4.3	3.5 / 6.8	5.5 / 10.1	7.2 / 12.7	9 / 15.5	2.2 / 4.3	3.5 / 6.8	5.5 / 10.1	7.2 / 12.7	9 / 15.5
7	Simple	1226	1969	3079	4255	5237	1926	3094	4838	6686	8230
	Multiple	943	1516	2371	3091	3727	1482	2382	3726	4857	5857
	Min. Bearing	1.9 / 3.7	3 / 5.9	4.8 / 9.2	6.6 / 11.9	8.1 / 14.4	1.9 / 3.7	3 / 5.9	4.8 / 9.2	6.6 / 11.9	8.1 / 14.4
8	Simple	937	1505	2355	3275	4253	1472	2366	3700	5147	6683
	Multiple	721	1159	1813	2522	3089	1133	1821	2849	3963	4855
	Min. Bearing	1.7 / 3.2	2.7 / 5.1	4.2 / 8	5.8 / 11.1	7.5 / 13.6	1.7 / 3.2	2.7 / 5.1	4.2 / 8	5.8 / 11.1	7.5 / 13.6
9	Simple	739	1188	1858	2585	3379	1161	1867	2920	4063	5309
	Multiple	568	914	1430	1990	2601	893	1436	2248	3127	4088
	Min. Bearing	1.5 / 3	2.4 / 4.6	3.7 / 7.1	5.1 / 9.9	6.7 / 12.9	1.5 / 3	2.4 / 4.6	3.7 / 7.1	5.1 / 9.9	6.7 / 12.9
10	Simple	597	961	1503	2092	2734	939	1510	2363	3287	4296
	Multiple	459	739	1157	1610	2104	721	1161	1818	2530	3307
	Min. Bearing	1.5 / 3	2.1 / 4.1	3.3 / 6.4	4.6 / 8.9	6 / 11.6	1.5 / 3	2.1 / 4.1	3.3 / 6.4	4.6 / 8.9	6 / 11.6
11	Simple	487	792	1241	1727	2257	765	1245	1950	2713	3547
	Multiple	378	609	954	1328	1737	594	957	1499	2087	2729
	Min. Bearing	1.5 / 3	1.9 / 3.7	3 / 5.8	4.2 / 8.1	5.5 / 10.6	1.5 / 3	1.9 / 3.7	3 / 5.8	4.2 / 8.1	5.5 / 10.6
12	Simple	373	665	1041	1449	1895	587	1044	1636	2277	2977
	Multiple	317	510	800	1114	1457	498	802	1257	1751	2290
	Min. Bearing	1.5 / 3	1.8 / 3.4	2.8 / 5.3	3.9 / 7.4	5 / 9.7	1.5 / 3	1.8 / 3.4	2.8 / 5.3	3.9 / 7.4	5 / 9.7
13	Simple	292	565	886	1233	1612	459	888	1392	1937	2534
	Multiple	269	434	680	948	1240	423	682	1069	1489	1948
	Min. Bearing	1.5 / 3	1.6 / 3.2	2.6 / 4.9	3.6 / 6.9	4.6 / 9	1.5 / 3	1.6 / 3.2	2.6 / 4.9	3.6 / 6.9	4.6 / 9
14	Simple	233	478	762	1061	1388	366	751	1198	1668	2182
	Multiple	231	373	585	815	1067	363	586	920	1281	1677
	Min. Bearing	1.5 / 3	1.5 / 3	2.4 / 4.6	3.3 / 6.4	4.3 / 8.3	1.5 / 3	1.5 / 3	2.4 / 4.6	3.3 / 6.4	4.3 / 8.3
15	Simple	188	387	663	923	1208	296	608	1041	1451	1893
	Multiple	200	324	508	709	928	315	509	799	1114	1455
	Min. Bearing	1.5 / 3	1.5 / 3	2.2 / 4.3	3.1 / 5.9	4 / 7.8	1.5 / 3	1.5 / 3	2.2 / 4.3	3.1 / 5.9	4 / 7.7

PROJECT ELEMENT PARAMETERS FOR ROOF

GLT roof purlins (beams):
3 ½” x 9 ½” Glulam section
(self-weight 8 plf)

GLT BEAM SPAN & LOAD TABLE

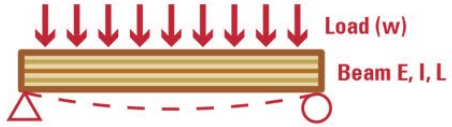
MPP Structural Floor Deck Single Span

MASS PLY BY FRERES

MPP panel width options:
3’-10”, 7’-10”, 9’-10”, or 11’-10”

FLOOR TABLE NOTES:

- Single Span, uniform load.
- Total load includes live load, Mass Ply weight, and 15 PSF dead load.
- Live Load deflection limit = Span (L)/360, total load deflection limit = L/240.
- Vibration controlled spans calculated using concepts outlined in the CLT handboook for a floor with no topping.



ENGINEERING OFFICE
50 psf Live Load: **< 8.92’ span**

GROCERY STORE
75 psf Live Load: **< 7.76’ span**

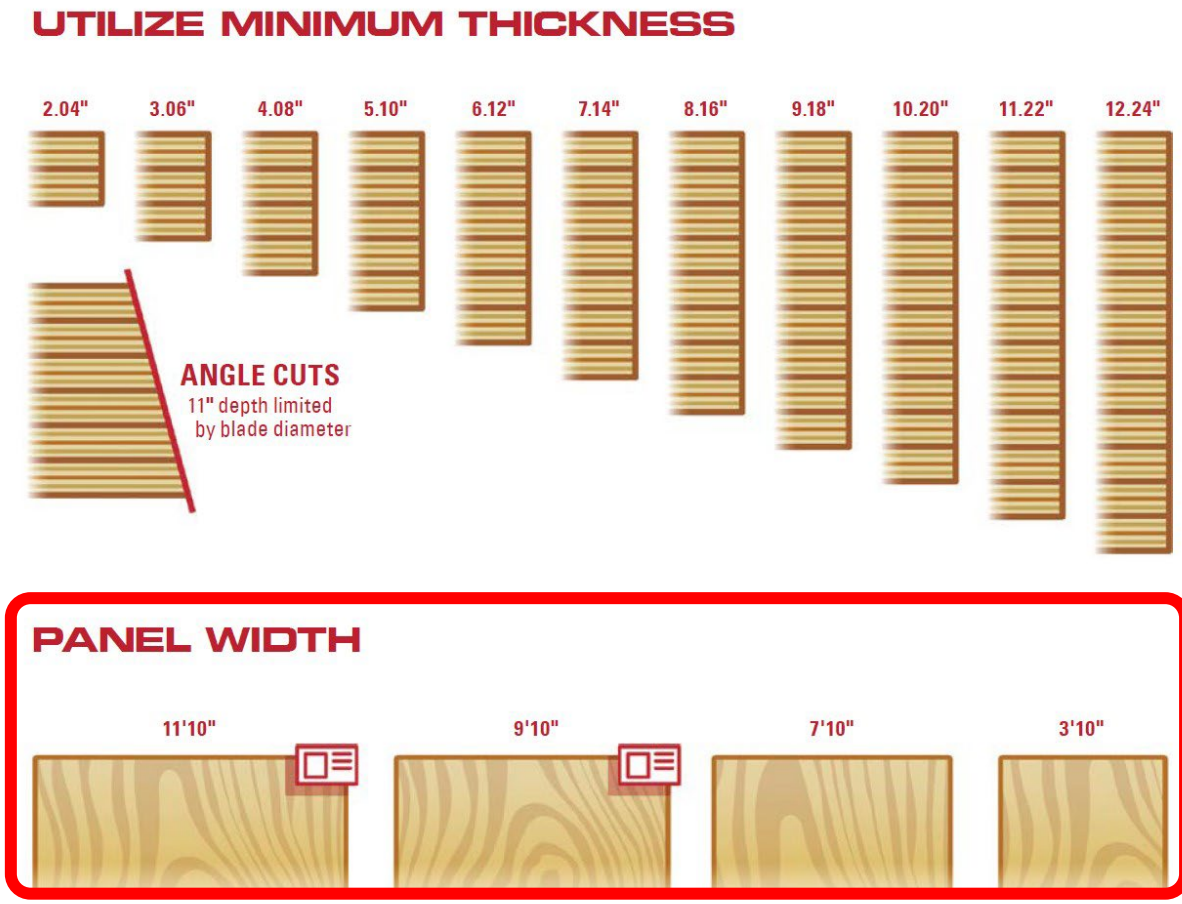
LIBRARY
150 psf Live Load: **< 6.09’ span**

F10 MASS PLY MAXIMUM SPAN FLOOR TABLE - MAJOR STRENGTH AXIS

LAY-UP	THICKNESS	VIBRATION CONTROLLED	LIVE LOAD				
ID	(IN)	SPAN (FT)	40 PSF	50 PSF	75 PSF	100 PSF	150 PSF
F10-2	2 1/16	6.20	6.36	5.94	5.17	4.68	4.07
F10-3	3 1/16	8.44	9.41	8.92	7.76	7.03	6.09
F10-4	4 1/16	10.41	12.28	11.71	10.30	9.33	8.10
F10-5	5 1/8	12.32	15.14	14.47	12.91	11.69	10.14
F10-6	6 1/8	14.12	17.90	17.14	15.48	14.02	12.17
F10-7	7 1/8	15.83	20.58	19.74	18.05	16.35	14.20
F10-8	8 3/16	17.50	23.21	22.29	20.49	18.68	16.22
F10-9	9 3/16	19.10	25.78	24.80	22.85	21.03	18.26
F10-10	10 3/16	20.66	28.28	27.23	25.15	23.36	20.29
F10-11	11 1/4	22.19	30.77	29.64	27.43	25.70	22.32
F10-12	12 1/4	23.67	33.14	31.99	29.67	27.89	24.34

MPP SPAN TABLE…3” thick panel (F10-3); span determined by Live Load

MPP standard panel widths





MPP panel 3-inch thickness and width options of 11’-10”, 9’-10”, 7’-10”, and 3’-10”

MPP FLOOR DECK ***…optimized widths for efficient prefabrication process & design options***